ENGINEERING PORTFOLIO / 2025–26

# STREETVAULT

## A hydraulically operated, multi-layer underground parkin

Reimagining residential road space through mechanical design thinking.




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**Seeking a 2027 Industrial Placement**

01 / PROJECT SNAPSHOT

# An urban parking challenge — approached as a mechanical design problem

A team concept developed for Design and Practice at London South Bank University.

## THE CHALLENGE

Residential streets have competing demand

## THE CONCEPT

Store vehicles vertically below the road using

## THE DESIGN INTENT

Keep the road surface available for normal

## THE CONTEXT

A proposed response for dense residential areas in Lo

### PROJECT TEAM

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### ACADEMIC CONTEXT

Design and Practice
School of Engineering
Submitted: 19 December 2025

### MY PORTFOLIO FOCUS

Problem framing • concept development
CAD exploration • system thinking
Risk and feasibility awareness

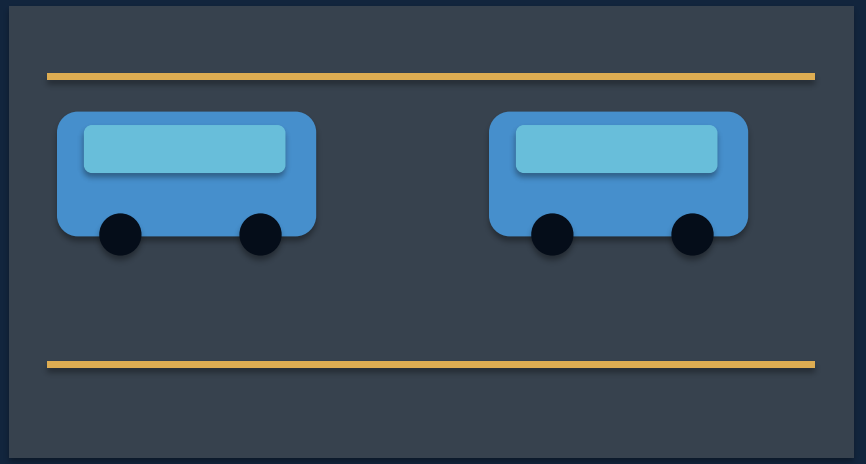
02 / PROBLEM DEFINITION

# Residential roads are being asked to do too many jobs

The report identifies the tension between long-term vehicle storage and the road's role as a public movement corridor.

- Parked vehicles reduce effective carriageway width and can make two-w
- Restricted access can complicate the passage of emergency vehicles.
- Parking partly on pavements can create barriers for pedestrians, wheel
- Reduced lateral clearance can make conditions less comfortable for cy

THE SPACE CONFLICT

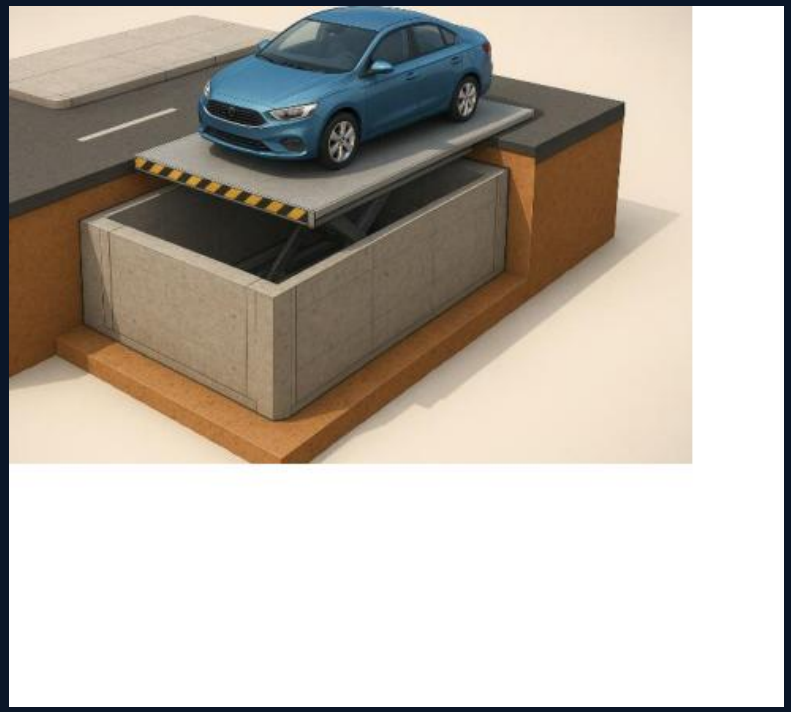


*Illustrative problem framing — not measured traffic data.*

03 / DESIGN BRIEF

# Move parking underground; return the surface to the street

The proposal uses the footprint of a roadside parking space to accommodate vertically stacked vehicle platforms.



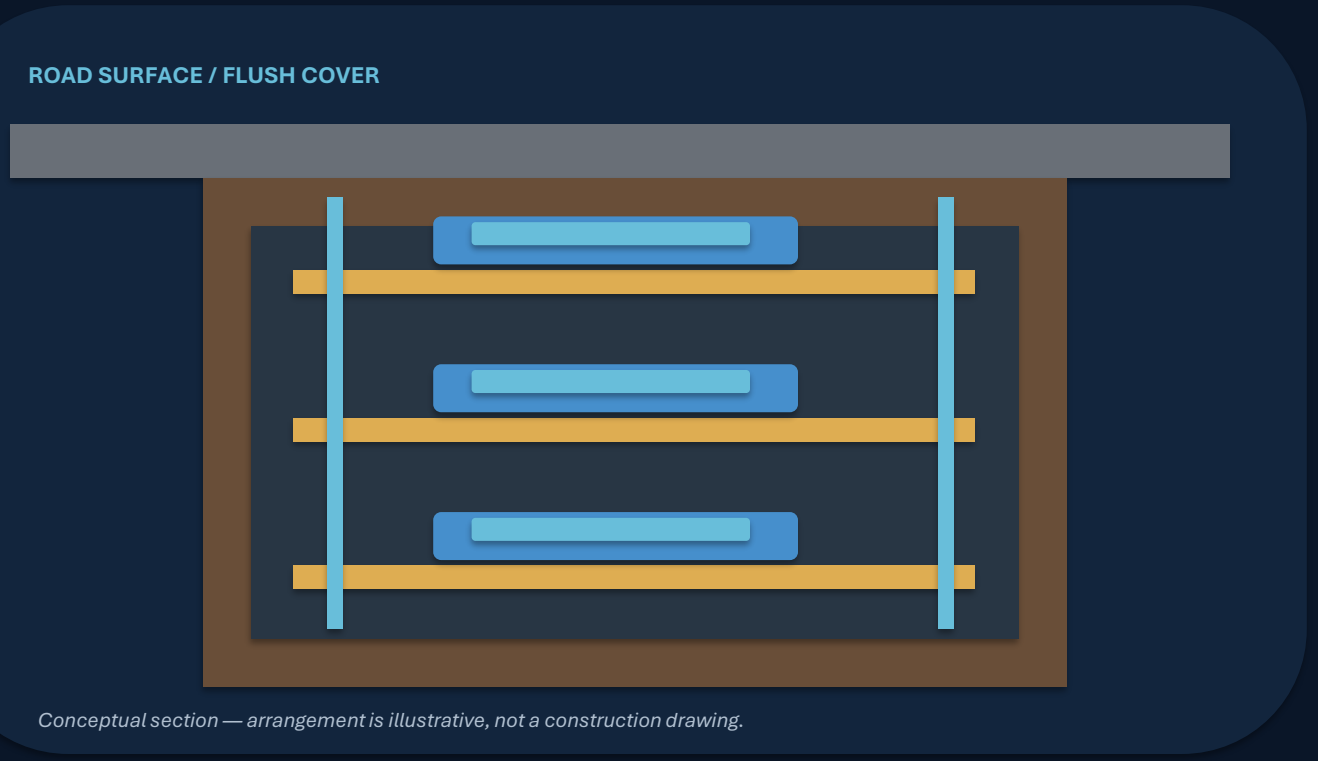
- Reduce long-term roadside parking pressure.
- Restore more of the road width for public movement when the system is idle.
- Improve pedestrian accessibility and emergency access potential.
- Use modular vaults that could be considered for phased deployment.
- Consider future integration with controlled access and EV charging.

**Design ambition: increase parking capacity per surface footprint without permanently occupying the carriageway.**

04 / CONCEPT ARCHITECTURE

# A layered system: road surface, lifting platform and underground vault

The report proposes two to three vertically arranged vehicle positions within a single vault footprint.




*Conceptual section — arrangement is illustrative, not a construction drawing.*

### 01 ROAD DECK

Flush surface intended to carry normal street use when closed.

### 02 LIFT PLATFORMS

Vertically stacked platforms carry vehicles between levels.

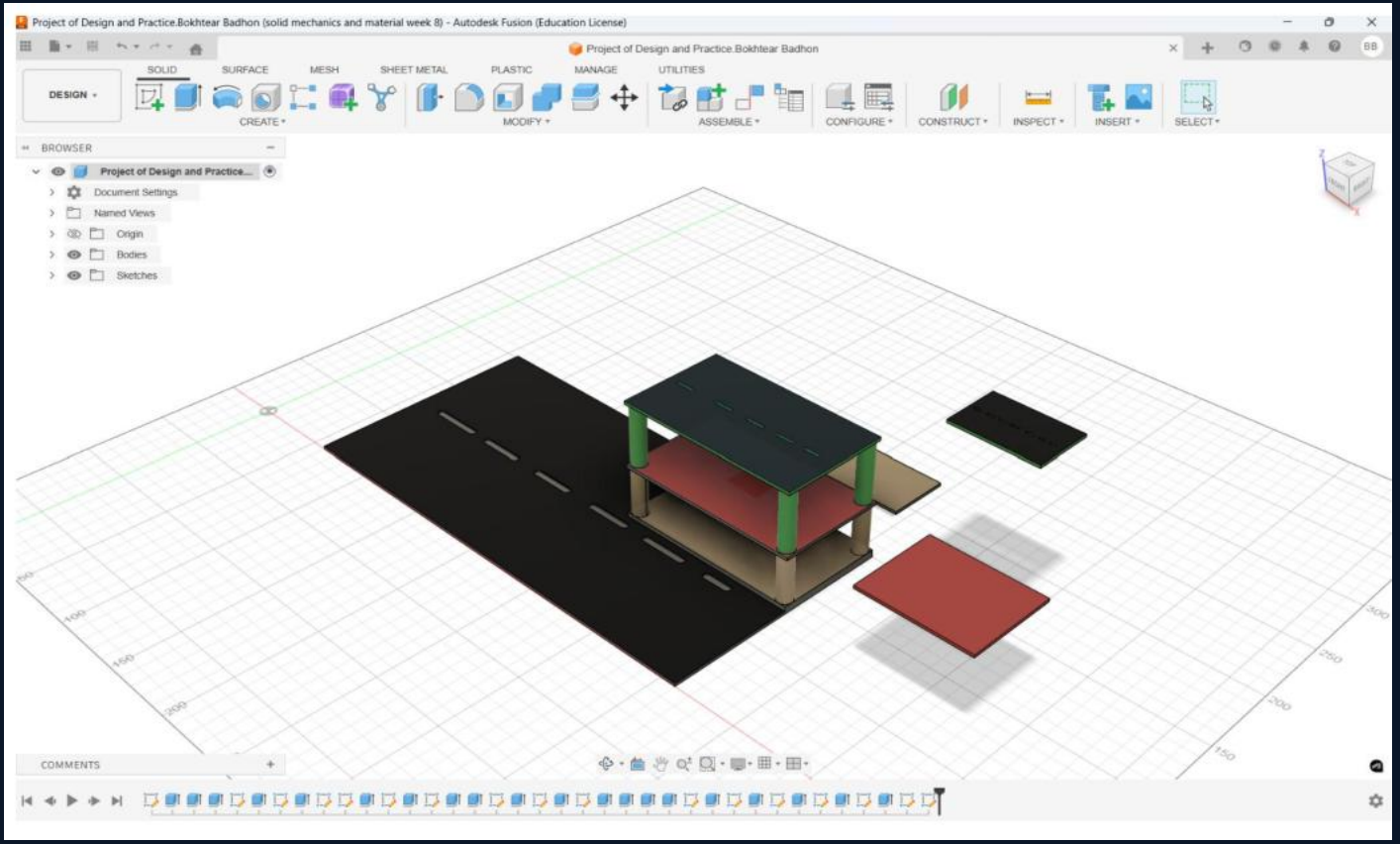
### 03 VAULT STRUCTURE

Reinforced-concrete enclosure with waterproofing and drainage provisions.

05 / DESIGN DEVELOPMENT

# Translating the concept into a Fusion 360 assembly

The submitted report includes an early CAD model showing the road plane, platform arrangement and supporting structure.




## WHAT THE MODEL COMMUNICATES

- Overall spatial arrangement
- Relationship between the road plane and vault
- Platform and support components
- An early stage of design iteration

### NEXT CAD STEP

Add dimensions, clearances and detailed interfaces.

06 / MECHANICAL PRINCIPLE

# Hydraulics provide controlled vertical movement

The report selects hydraulic actuation because it can generate high lifting force in a compact arrangement.

**ELECTRIC MOTOR**
Drives the pump

›

**HYDRAULIC PUMP**
Pressurises fluid

›

**CONTROL VALVES**
Manage flow and direction

›

**CYLINDERS**
Convert pressure to linear motion

›

**PLATFORM**
Raises / lowers the vehicle

**ENGINEERING FOUNDATION**

Pascal’s Law: pressure applied to a confined fluid is transmitted through the fluid, enabling force multiplication at the act

*The report proposes four synchronised cylinders per platform to support load distribution and reduce uneven lifting.*

07 / OPERATING WORKFLOW

# A controlled sequence is central to safe public-road operation

The following sequence is adapted from the parking and retrieval protocols in the report.

**01 REQUEST**

User initiates a parking request through the control inter

**02 CLEAR ZONE**

Traffic / obstacle sensors confirm the operating area is c

**03 WARN & ISOLATE**

Warning signs and barriers activate before movement.

**04 RAISE PLATFORM**

Hydraulic system brings the selected platform to road le

**05 POSITION VEHICLE**

Driver enters; alignment and weight sensors check plac

**06 LOWER & SECURE**

Platform descends into the vault and is mechanically secu

**Retrieval reverses the sequence: authenticate → raise the selected platform → drive away → retract and secure.**

08 / SAFETY ENGINEERING

# Public infrastructure demands layered protection

The report identifies technical, structural, environmental and operational hazards — and proposes mitigation measures.

### HYDRAULIC FAILURE

Redundant cylinders • fail-safe valves • mechanical lockin

### UNCONTROLLED MOVEMENT

Position / pressure sensing • emergency stop • controlled

### WATER INGRESS

Waterproof membranes • drainage • sump pumping • inspection

### GROUND / SETTLEMENT

Geotechnical assessment • reinforced foundation design

### PUBLIC INTERACTION

Access control • automatic barriers • warning signs • user

### MAINTENANCE

Cylinder / seal checks • fluid filtration • sensor calibration • struc

*These are proposed safeguards in the concept report; they have not been demonstrated through physical testing.*

09 / FEASIBILITY & CONSTRAINTS

# A promising concept — with substantial implementation questions

The report recognises that underground construction introduces cost, utility, environmental and approval challenges.

## OPPORTUNITIES

- Several vehicle positions within one surface-space footprint.
- Potentially clearer carriageways and pavements.
- Modular deployment could support pilot schemes.
- Possible future links to smart access and EV charging.

## CHALLENGES TO RESOLVE

- High excavation and reinforced-concrete capital costs.
- Existing underground utilities and ground conditions.
- Groundwater, flooding, drainage and waterproofing.
- Planning approvals, liability, disruption and public acceptance.

*No quantified cost model, geotechnical survey or full structural verification is included in the submitted report.*

10 / INTENDED IMPACT

# Designed around the street as a shared public space

The project evaluates potential benefits for residents, road users, local authorities and the urban environment.

### RESIDENTS

More predictable parking access and redu

### PEDESTRIANS & CYCLISTS

Potentially clearer pavements, improved va

### EMERGENCY SERVICES

A less obstructed carriageway could suppo

### LOCAL AUTHORITIES

Potential alignment with street accessibility and ma

**Sustainability proposition: use vertical underground space to preserve surface land for movement, greenery and social use.**

11 / ENGINEERING DEVELOPMENT

# What this project helped me practise

A first-year design project is an opportunity to connect engineering fundamentals with a complex real-world need.

### PROBLEM DEFINITION

Breaking a broad urban issue into functional design objecti

### MECHANICAL THINKING

Exploring hydraulic actuation, force transmission and platf

### CAD COMMUNICATION

Using Fusion 360 and concept visuals to communicate spatial arr

### SYSTEMS THINKING

Considering sensors, controls, structures, drainage and us

### RISK AWARENESS

Recognising failure modes, public safety and maintenance

### CRITICAL EVALUATION

Identifying what still needs calculations, testing and technical val

**Next development priorities: detailed dimensions • load calculations • hydraulic sizing • ground investigation • prototype / simulation testing**



# BOKHTEAR UDDIN BADHON

**BEng (Hons) Mechanical Engineering | London South Bank University**

Curious about how engineering ideas become practical, safe and useful systems.

**CURRENT FOCUS**

Mechanical design • CAD development • fluid power • sustainable engineering • technical communication

**CAREER OBJECTIVE**

## Seeking a 2027 Industrial Placement in London

Let’s connect: LinkedIn | Portfolio | S4431048@lsbu.ac.uk